

TEAM NAME- GKG

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PROBLEM STATEMENT

Amazon loses **billions of dollars annually** to fraud, but the real cost is customer trust. Current fraud detection works in isolation - catching individual fake reviews or single counterfeit products. Meanwhile, organized fraud networks operate freely, using sophisticated tactics:

- **Review Farms** : 1000+ fake accounts writing reviews for hundreds of sellers
- **Counterfeit Rings** : Networks selling fake products across multiple storefronts
- **Coordinated Attacks** : Competitors destroying legitimate sellers with review bombs
- **False Positives** : Current systems harm legitimate sellers with incorrect fraud flags

The Hidden Truth : From manually investigating fraud examples on Amazon, we've observed that fraudulent activities can be connected or isolated. One fake reviewer can review dozens of products. One fraudulent seller typically operates multiple storefronts. Current systems treat each incident separately and lack proper verification mechanisms.

SCOPE AND INNOVATION

Our Innovation: TRUSTSIGHT - Cross Intelligence with Decision Memory

TrustSight introduces two innovative concepts:

1. **Cross Intelligence** : Every fraud signal becomes a network investigation. When we detect a review, instead of just removing it, we:
    a. Trace the reviewer's entire history
    b. Identify all products they've reviewed
    c. Investigate those sellers
    d. Map connections to other reviewers
    e. Expose the entire fraud network
2. **Fraud Decision Memory Bank** : Every detection is tracked, verified, and learned from. When we detect potential fraude, we:
    a. Record the decision with all evidence
    b. Track verification outcomes
    c. Only fine-tune models on verified fraud
    d. Learn from false positives to prevent repeats
    e. Provide appeals mechanism for sellers for reducing system's false positives

Result : One fake review leads to shutting down entire fraud rings worth millions, while protecting legitimate sellers from false accusations.

Scope

- **Counterfeit Detection:** Combines image analysis, pricing, and seller behavior. Flags stock-only or AI-generated images; micro-verifies mixed image sources.
- **Review Authenticity:** Uses LLMs to assess genuineness and detect patterns across reviewer networks.
- **Network Mapping:** Graph-based tracing links related reviews, products, and sellers once any signal is found.
- **Lifecycle Monitoring:** Follows products from listing to return to identify abuse.
- **Predictive Prevention:** Forecasts future fraud trends for soft warnings (no hard action).
- **Decision Verification:** All actions are tracked, verified, and appealable before severe steps like bans.

WORKING BACKWARDS FROM CUSTOMER

Primary Customer: Amazon Shoppers

Current Experience :

- "Is this Nike shoe real or fake?"
- "Can I trust these reviews?"
- "Why did I receive a counterfeit?"

With TrustSight :

- See a simple Trust Score (0-100) on every product
- **94% score** = Buy with confidence
- **45% score** = Clear warning with reasons
- One number answers: "Can I trust this?"
- Report suspicious products easily

Secondary Customer: Legitimate Sellers

Current Pain :

- Lose sales to counterfeiters selling at impossible prices
- Suffer review bombs from competitors
- Get falsely flagged as fraudulent with no recourse
- No warning before attacks

With TrustSight :

- Get alerts: "Potential review attack detected - monitoring increased"
- See fraudulent competitors removed in bulk
- **Appeal false positives** through Memory Bank system
- Protected from incorrect fraud flags
- Fair marketplace where quality wins

Tertiary Customer: Amazon Teams

Current Process :

- Assuming Amazon does manual investigation of individual reports
- Miss network connections
- No way to track decision accuracy
- Reactive, not proactive

With TrustSight :

- One-click fraud network visualization
- See how 1 fake review connects to 1000s more
- Track false positive rates and improve
- Proactive prevention saves millions
- Complete audit trail of all decisions

TARGET PERFORMANCE METRICS

Detection Accuracy (after verification)

True Positive Rate
False Positive Rate
Network Discovery rate

Verification Effectiveness

Decisions verified within one week
False positives caught after verification
Average verification time
Seller appeal success rate

Learning Effectiveness

Model improvement from feedback
False positive reduction over time
New patterns discovered monthly

Prediction Accuracy

Review bomb prediction
Exit scam prediction

Operational Efficiency

Processing Speed
Real Time latency
Cost per check

Business Impact

Fraud prevented annually
Legitimate Sellers protected
Customer complaints reduced

IMPACT OF OUR SOLUTION

What Success Looks Like

Goal: Detect more fraud by uncovering connected activities.

Method: Convert each fraud signal into a network investigation.

Verification: Measure accuracy via the Decision Memory Bank.

Why It Works:

Fraudsters work in groups—linking accounts, products, and reviews. Current systems miss these ties. TrustSight maps relationships to expose entire fraud rings, not just individuals.

Impact:

- Preventing just 1% of fraud = millions saved
- 50% fewer false positives = thousands of sellers protected
- 10% fewer fraud hits = stronger marketplace trust

Our Thinking

Fraud is networked—so detection should be too. TrustSight pioneers **Cross Intelligence** , tracing connections across accounts, reviews, and products.

Our **Decision Memory Bank** ensures constant learning from both hits and misses.

Even conservative gains = major impact:

- 1 fraud lead → 10 discoveries
- Preventing 1% of fraud = millions saved
- 100 sellers protected = stronger trust

The Innovation: We're the first to map all fraud signals as networks and track accuracy over time. Metrics will follow—but the principle is clear: fraud is connected, and detection must evolve to match.

SCALABILITY AND AMAZON ECOSYSTEM EXPANSION

Scalability Design

Smart Prioritization

- Not every product needs deep analysis
- AI prioritizes based on risk factors:
  - **HIGH**: Luxury items, new sellers, anomalies (Deep analysis)
  - **MEDIUM**: Unusual patterns detected (Standard analysis)
  - **LOW**: Established sellers, normal patterns (Light monitoring)
  - **SKIP**: Digital goods, books <$20 (Minimal resources)

Technical Scalability

- Cloud-native architecture using AWS
- Designed to handle Amazon's scale
- Efficient caching and smart sampling
- Cost-effective per-analysis pricing
- Decision Memory Bank removes old verified fraud caught over the time and follows a tiered architecture

Expansion Roadmap

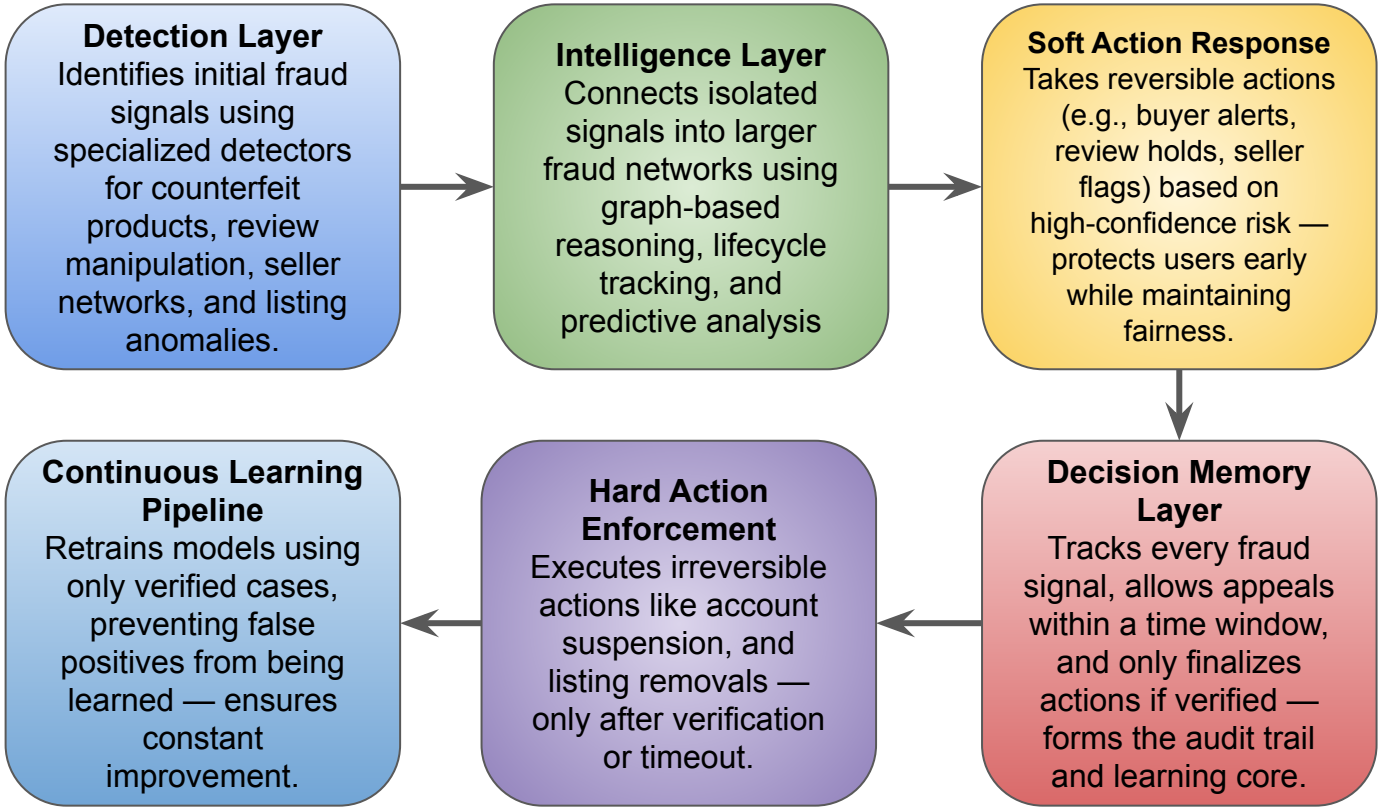
Phase 1: Core Marketplace (Months 1-6)

- Start with high-fraud categories (Electronics, Fashion, Shoes)
- Expand to all physical products >$20
- Cover all major product categories
- Build Decision Memory Bank foundation

Phase 2: Amazon Ecosystem (Months 6-12)

- **Amazon Business** : B2B fraud detection
- **Amazon Fresh/Whole Foods** : Food authenticity and supply chain verification
- **Amazon Pharmacy** : Medication authenticity and seller verification
- **Prime Exclusive Deals** : Ensure deal integrity
- Expand verification workflows

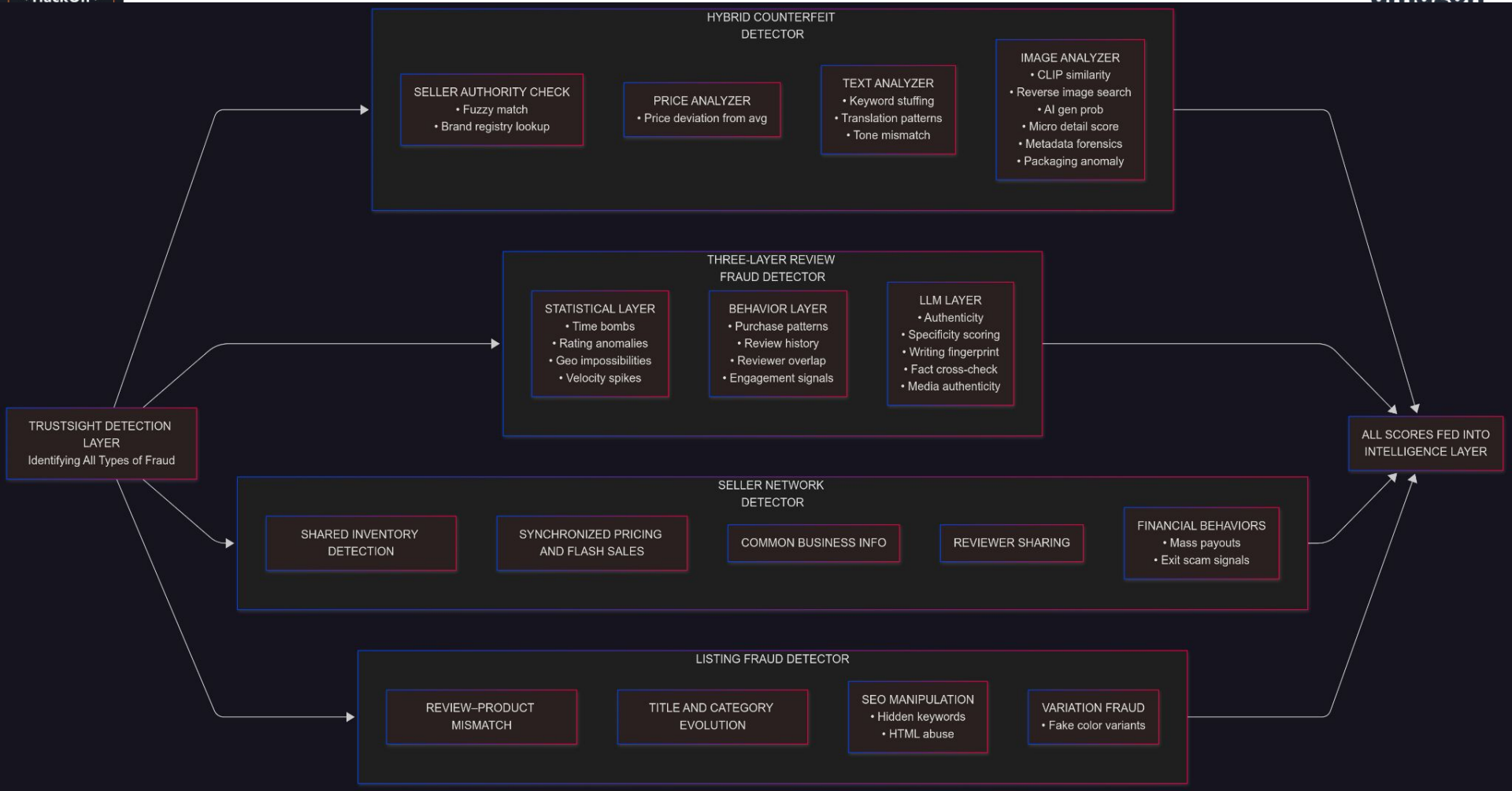
ARCHITECTURE



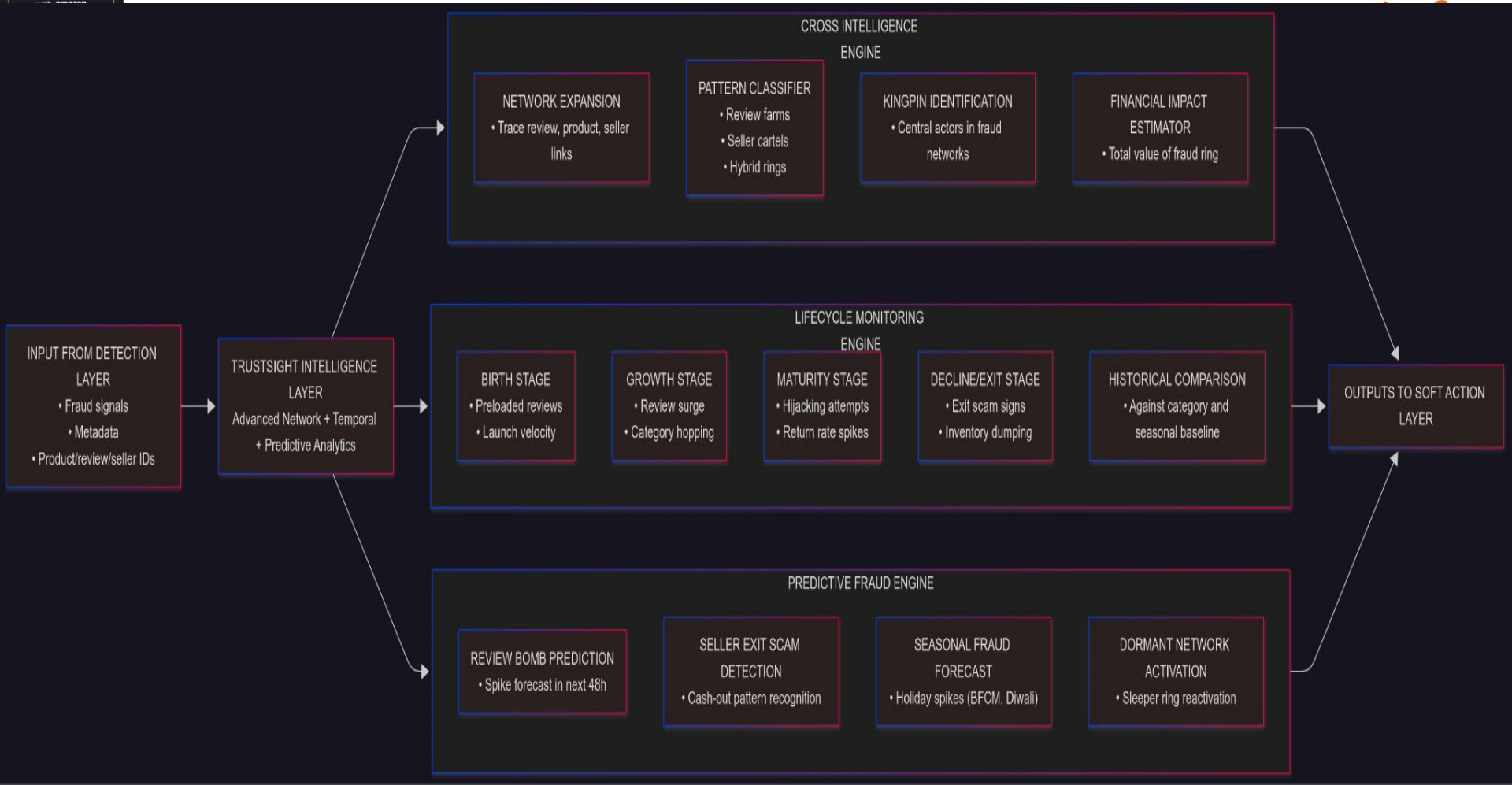
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DETECTION LAYER (LINK)

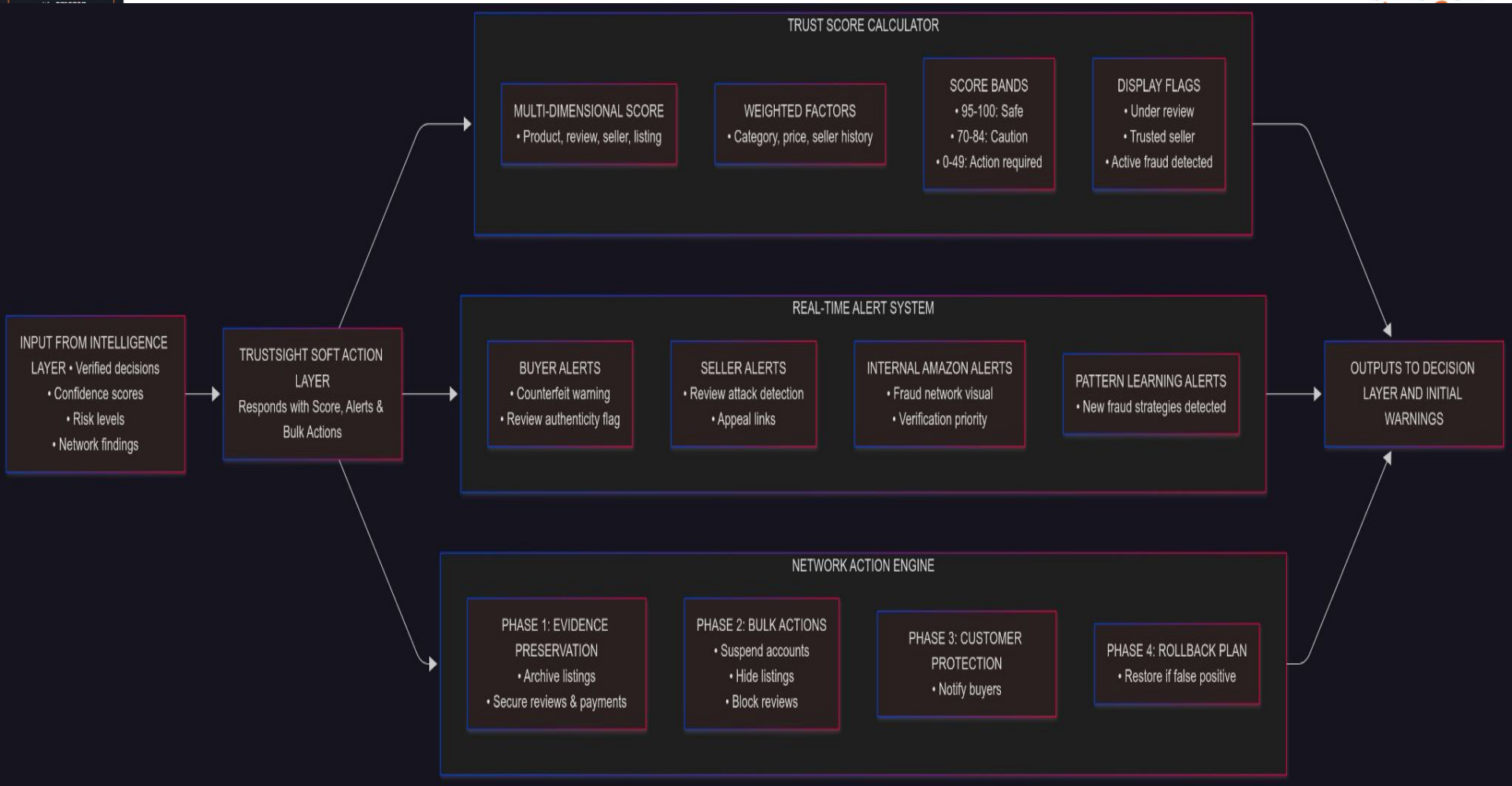
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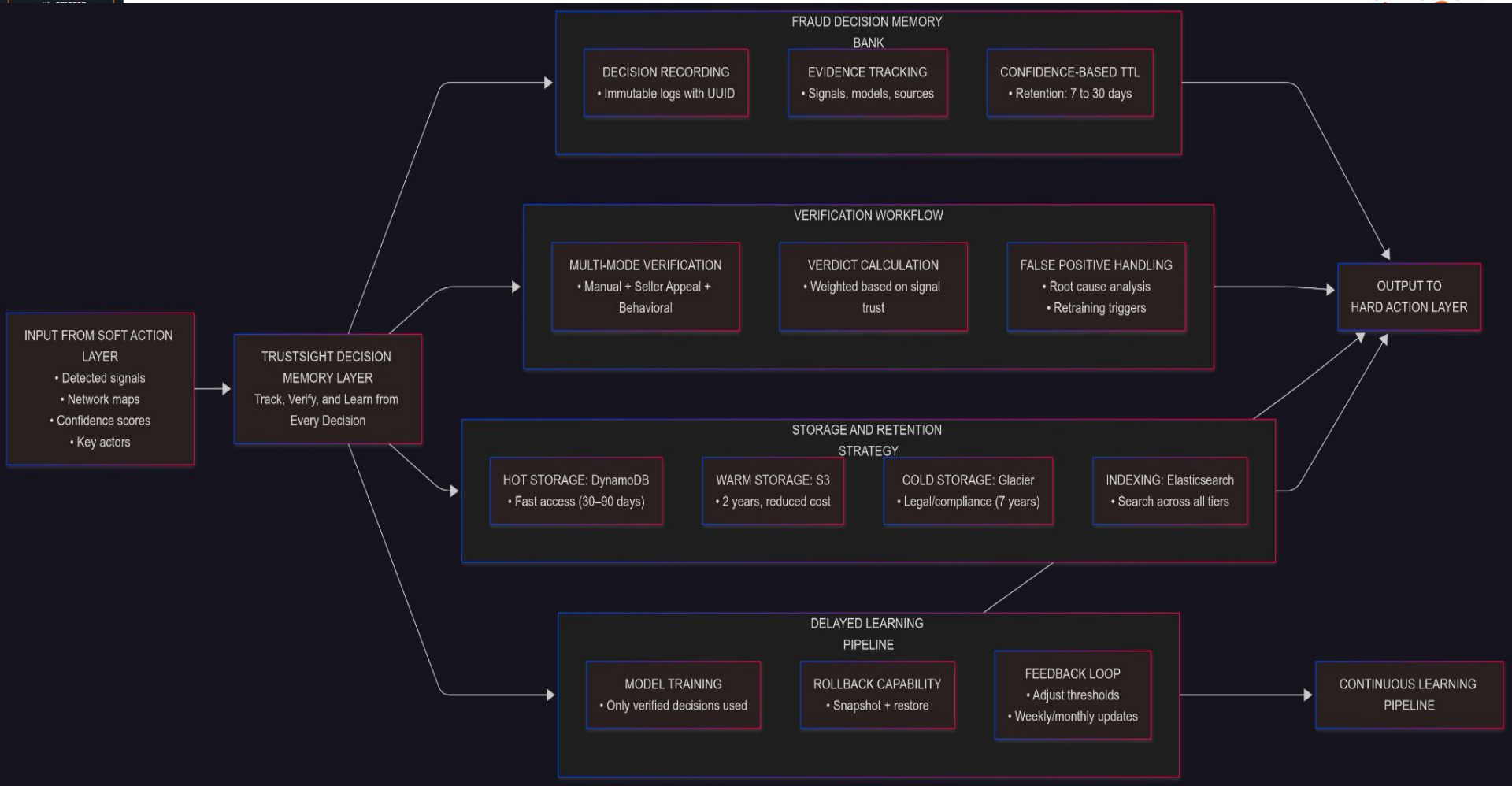
INTELLIGENCE LAYER (LINK)



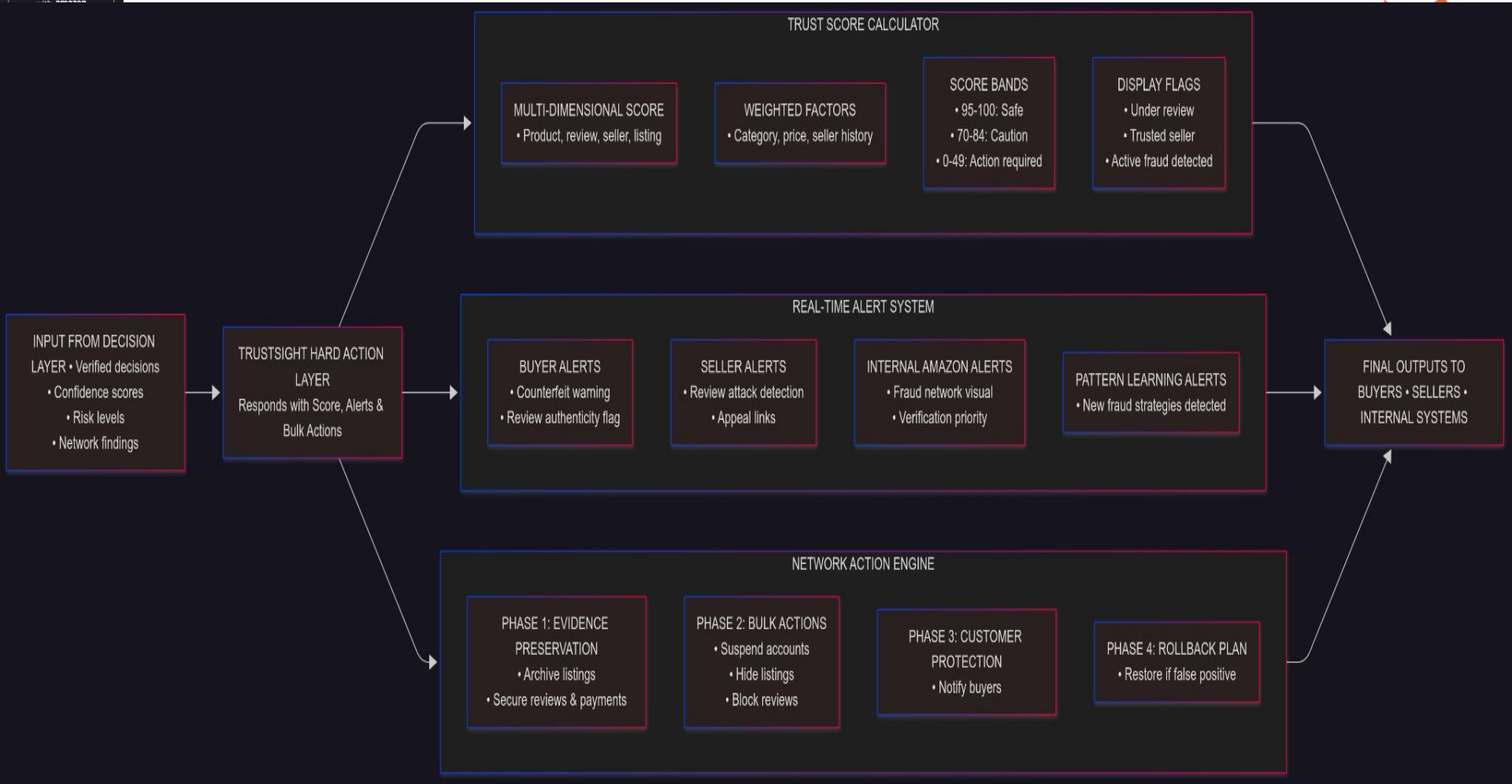
SOFT ACTION LAYER (LINK)



DECISION MEMORY LAYER (LINK)



HARD ACTION LAYER (LINK)



ADDITIONAL KEY FEATURES

Continuous Improvement

Unlike static systems, TrustSight gets smarter:

- Every verified fraud improves detection
- Every false positive prevents future mistakes
- Every appeal teaches the system
- Every day brings better accuracy

Multi-Layer Protection:

1. **Detection Layer** catches individual signals
2. **Cross Intelligence** exposes hidden networks
3. **Lifecycle Monitoring** tracks temporal patterns
4. **Decision Memory** verifies and improves
5. **Predictive Engine** prevents future fraud

Risk Mitigation

- **False positives?** → Decision Memory Bank allows appeals
- **Wrong decisions?** → Verification workflow catches mistakes
- **Model degradation?** → Only trains on verified data
- **Seller concerns?** → Full transparency and appeal rights

GENERAL CASES COVERED

Counterfeit Detection - General Cases

✓ **Image-based fraud:**

- Stolen official brand images by unauthorized sellers
- AI-generated product photos
- Stock photos from wholesale sites (Alibaba, DHgate)
- Photoshopped/edited official images
- Images with removed watermarks or copyright

✓ **Price-based fraud:**

- Impossible pricing (Nike shoes $29 when retail $129)
- Synchronized price manipulation across sellers
- Loss-leader pricing before exit scams

✓ **Seller-based fraud:**

- Brand impersonation (Nikee vs Nike)
- Unauthorized sellers claiming official status
- New accounts selling high-value brands

Edge Cases Handled:

- Generic/private label products → Falls back to review/seller patterns
- Authorized resellers using official images → Authority database prevents false flags
- Regional pricing differences → Seller location verification
- Liquidation/clearance sales → Historical seller behavior check

Review Fraud - General Cases

✓ **Timing anomalies:**

- Reviews posted 3 minutes after delivery (impossible!)
- Reviews before product delivery
- All reviews posted 3-4 AM (bot timing)
- Review velocity spikes (50+ in 1 hour)

✓ **Pattern detection:**

- Group behavior (10 accounts reviewing in sequence)
- Same reviewers always appearing together
- Template/copied reviews across products
- Rating-text mismatches ("Terrible product! 5 stars")

✓ **Geographic fraud:**

- Same reviewer in NYC and LA same day
- All reviewers from one zip code
- VPN usage patterns
- Shipping address mismatches

✓ **Content fraud:**

- Generic praise without specifics
- AI-generated review text
- Fake review media (stock photos as "customer photos")
- Reviews mentioning wrong product features

Edge Cases Handled:

- Beta tester reviews at launch → Reviewer history verification
- Incentivized reviews → Language pattern detection
- Cultural language differences → Multi-language LLM analysis

Other cases covered are

-Seller network fraud patterns including price coordination (multiple sellers dropping prices simultaneously), inventory sharing across "different" sellers, registration clusters, and exit scam preparations.

-We detect listing fraud such as hijacking (seller/location changes), category gaming (shoes listed as books), SEO manipulation, and variation abuse.

-Our lifecycle monitoring catches birth anomalies (50 reviews in week one), growth manipulation through review bombs, and decline-stage exit scams.

-Special sophisticated patterns include time-delayed fraud (6 months legitimate then fraudulent), brushing scams for fake metrics, wardrobing networks (buy-use-return schemes), review laundering (mixing fake with real), sleeper cell activations, and competitor sabotage.

-Edge cases like legitimate wholesale operations, franchises, seasonal businesses where purchase rate go high, viral products, and limited editions are distinguished from fraud through business verification, authority checks and historical patterns.

Supportive Fallback

The multi-layered system ensures if image detection misses something, review patterns catch it; if reviews miss it, network connections reveal it; if networks seem clean, lifecycle anomalies show it; and if we make any mistakes, the Decision Memory Bank corrects them through verification and continuous learning.